CS490 Final Project Report

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Abstract

Toxicity detection in online gaming is primarily done through binary classification that confuse genuine harassment with competitive banter. To try to resolve this, we fine-tune a pretrained language model to perform finer grained toxicity classification across three categories: toxic, competitive banter, and neutral. By training on gaming-domain chat data with this new label, we aim to reduce false positives on competitive speech while maintaining strong detection of genuine harassment. We evaluate against binary toxicity baselines to see if that label granularity meaningfully improves both classification accuracy and alignment with community communicative norms.

1 Introduction

Few online spaces have developed communicative norms quite like those of multiplayer video games; specifically, those that allow in-game communication between players competing against one another. Unlike most other forms of social media, gaming chat is characterized by high-stakes competition, real-time interaction, and a culture of talking trash that has become widely accepted, even expected by the community. Phrases that could be labeled as hurtful in other online settings carry very different contexts in a post-game lobby. With advancing Natural Language Processing models, much research has been done in attempting to develop contemporary models that can identify toxic speech, preventing offensive or excessive verbiage from being expressed.

Existing approaches to automated toxicity detection have enabled safer online gaming environments and the moderation of offensive users. However, many users and gamers would argue there is a flaw with this approach. Most current systems treat toxicity as a one-dimensional problem: a message is either toxic or it isn't, or it is scored along a single severity axis. That leaves no room for "competitive banter". Such systems classify these expressions as "toxic" or "netural", failing to capture the nuance that distinguishes competitive banter from actual harassment or hate speech. This results in the over-moderation of the community, with players banned for banter and the online gaming experience itself degraded in the process.

In this work, we build a tri-class classifier that distinguishes between three categories of in-game speech:

- **Neutral** — coordination, callouts, and flat reactions with no aggressive tone (e.g., "mid mia", "gg wp", "need wards").
- **Banter** — speech that is aggressive in tone but bounded by the game: trash talk, salty venting, gameplay-directed insults, and crude humor with no real-world target (e.g., "ez", "uninstall", "trash offlane").
- **Toxic** — targeted harassment, slurs, hate speech, sexualized threats, and attacks on real-world identity that would remain hostile outside the game context.

By introducing banter as its own class, we aim to preserve what many consider an important part of video game communication while still filtering out overly crude, offensive, targeted, or hate-speech riddled messages.

2 Related Work

The most widely used tool for toxicity detection is the Perspective API developed by Jigsaw and Google, which scores text on a continuous toxicity scale but reduces the classification problem to a single dimension. Davidson et al. 2017 represent an important step toward finer-grained classification, distinguishing between hate speech, offensive language, and neither in a Twitter corpus.

This three-class formulation is structurally similar to ours. Their finding that offensive language is frequently mislabeled as hate speech parallels our central concern about competitive banter being mislabeled as toxic. Founta et al. 2018 extend this further with a large-scale Twitter dataset annotated across multiple abuse categories, demonstrating that label granularity significantly affects model behavior and downstream utility.

Märtens et al. 2015 provide one of the earliest quantitative studies of toxic behavior in online gaming, analyzing player reports in League of Legends and identifying behavioral patterns associated with toxicity. Subsequent work has examined chat logs from games including Dota 2 and Counter-Strike, finding that domain-specific slang, abbreviations, and context-dependent insults pose significant challenges for general-purpose classifiers trained on social media data. These studies motivate the need for domain-adapted models rather than direct transfer from Twitter or Reddit corpora.

The introduction of BERT (Devlin et al., 2019) established fine-tuning pretrained transformers as the dominant paradigm for text classification tasks. We build on this line of work by fine-tuning a BERT-large model on gaming-domain data with our three-way label model.

3 Method/Approach

3.1 Dataset

We sourced our data from the DOTA 2 Toxic Chat Dataset on Hugging Face, which originally contained 2,500 samples of raw in-game chat messages. Known to be one of the most toxic communities across the gaming world, it allows us to counteract the small dataset by being full of extreme, repetitive examples specific and niche to DOTA 2. Because the dataset only included binary toxicity labels, we needed to re-annotate every message under our three-class scheme (neutral, banter, toxic). To do this efficiently at scale, we assigned one of the three labels to each message manually, with each groupmate attempting their own classification to get a reliable annotation. Afterwards, we took any decisions we spent over 5-seconds on and confirmed with GPT-4o for more robust sampling. During this process, we removed 48 samples that were either in non-English languages or otherwise ineligible for meaningful classification, bringing our final dataset to 2,452 usable samples.

This change from our proposal — the only change — was much needed as those data points would serve as nothing but noise, and the data in other languages could not be reliably annotated by hand without knowing cultural norms and intentions. The resulting class distribution skews heavily toward neutral messages: 1,257 neutral (51.3%), 947 banter (38.6%), and 248 toxic (10.1%). This imbalance, particularly the small toxic class, is an important consideration that informed our use of weighted cross-entropy loss during training. We split the data into 1,962 training, 245 validation, and 245 test samples.

3.2 Modeling

To highlight the subtle difficulties in toxicity detection, we selected multiple model architectures to be trained on our dataset. By cross-analyzing each model's performance, we were able to test not only which model types performed best for our problem, but how optimizing each model led to better or worse performance.

The first model we built and tested was a Naive Bayes Classifier. The NBC was chosen to be a baseline because it is computationally efficient, fast to train, and has been proven to work well on text classification. While being able to achieve decent performance, we correctly assumed this model would exemplify where a statistical "word-counting" model fails compared to the LLMs we later built.

The second model we worked on was a plain BERT-base architecture, consisting of $\approx 110M$ parameters (Efimov, 2023). We wanted to use BERT due to its ability to understand intent and contextual intricacies; however, we wanted to leave space for optimizations. Thus, we modified the BERT-base to use Parameter-Efficient Fine-Tuning (Mangrulkar and Paul, 2023), specifically Low-Rank Adaptation (LoRA). We gathered metrics from both the model with and without LoRA to allow space to analyze how small enhancements affect model performance.

The final modeling approach was to build and train a BERT-large model $\approx 365M$ parameters (Efimov, 2023). We wanted to see how parameter size alone affected prediction performance. Thus, we utilized BERT-large + LoRA to isolate how parameter size enhances results while holding optimizations constant. Our assumption was that this final model would prove to be the highest-performing model.

We evaluate all models using precision, recall, and F1-score computed per class, along with overall accuracy. Because our dataset is imbalanced and our primary concern is catching toxic messages without over-penalizing banter, we use macro F1 as our main evaluation metric. Macro F1 weights all three classes equally regardless of size, which prevents the model from looking artificially strong by performing well only on the majority neutral class. During training, we save the best model checkpoint based on validation macro F1.

3.3 Training Details

All experiments were run on Google Colab using a T4 GPU, and every model converged within 20 minutes of training. For the Naive Bayes classifier, no additional tuning was needed beyond the default TF-IDF vectorization. For all BERT models, we tokenized inputs with a max sequence length of 128, which comfortably covers the length of typical in-game chat messages, and used a batch size of 16. We used the AdamW optimizer and weighted cross-entropy loss, where the class weights were computed inversely proportional to class frequency. This was necessary because the toxic class makes up only about 10% of the dataset, and without weighting, the model would have little incentive to learn to classify toxic messages correctly. The best model checkpoint was saved based on peak validation macro F1 to ensure we selected the version of the model that performed most evenly across all three classes, not just the majority class.

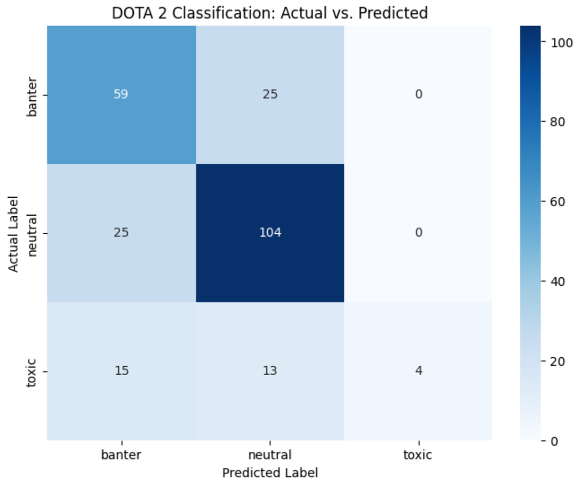
For BERT-base, we used a learning rate of 2e-5 and trained for 5 epochs with a full weight update across all parameters. Because LoRA freezes most of the model and only updates a small set of injected parameters, we increased the learning rate to 5e-5 and extended training to 10 epochs for BERT-base + LoRA, giving the smaller parameter budget more room to converge. The LoRA configuration used a rank of 8, alpha of 32, dropout of 0.1, and was applied to the query, key, value, and dense attention layers. For BERT-large + LoRA, to maintain the spirit of the experiment and findings, we kept the optimal convergence from BERT-base + LoRA, and just changed the model to BERT-large. This isolates the impact of adding the larger model.

4 Model Implementation and Results

4.1 Naive Bayes Classifier

NBC is a simple probabilistic classifier used on vectorized text. To convert our toxicity chats to usable data, we used Term Frequency-Inverse Document Frequency (TF-IDF) convert text into numerical vectors. This also scores words in the training set by importance, allowing validation and test messages to be numerical ratings.

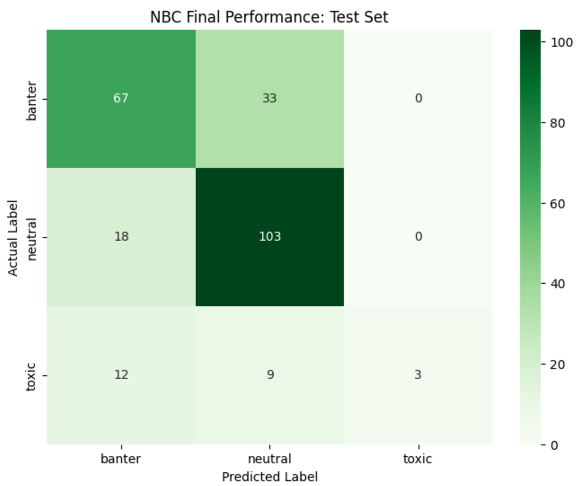
We trained the NB Model on the training set, then did predictions using the validation set. Without any fine-tuning on the vectorization process, we achieved the following metrics after training and validation, presented via confusion matrix:



Class	Precision	Recall	F1-score
banter	0.60	0.70	0.64
neutral	0.73	0.81	0.77
toxic	1.00	0.13	0.22
Accuracy	0.68	0.68	0.68
Macro Avg	0.78	0.54	0.54
Weighted Avg	0.72	0.68	0.65

Table 1: NBC performance, Training and Validation

Evaluating on the test set, we got the following change in results:



Class	Precision	Recall	F1-score
banter	0.69	0.67	0.68
neutral	0.71	0.85	0.77
toxic	1.00	0.13	0.22
Accuracy	0.71	0.71	0.71
Macro Avg	0.80	0.55	0.56
Weighted Avg	0.73	0.71	0.68

Table 2: NBC performance, Test Set

Based on these visuals, it is evident where the model is succeeding and where it is failing to capture the complexities required for proper toxicity classification.

The classifier seems to be doing well to predict neutral samples, but our dataset has mostly neutral samples. Statistical models like NBC have a tendency to push the most common label when in doubt, so even this seeming "success" is not a full credit to the model's abilities.

We see the model also does decently well classifying true "banter" labels as "banter". Where the model makes a mistake is in the 25 times it classified true "banter" labels as "neutral". However, given the context of why our project is important, this is a less harmful mistake to make; if it were classifying bantering messages as toxic, players could be wrongly penalized for their chat behavior.

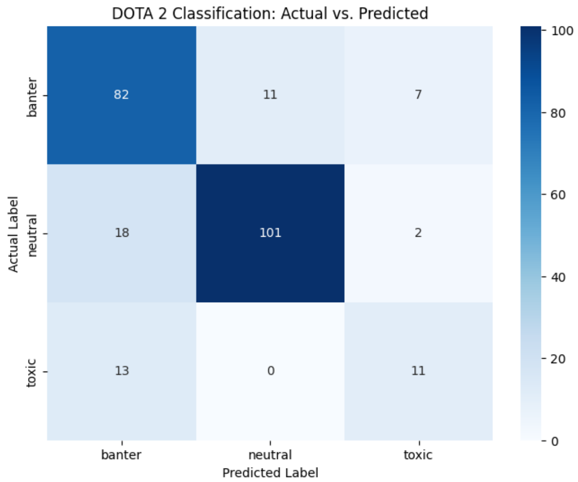
The true issue in this model lies in the cases of misidentifying toxicity. We see that only 4 out of the 32 toxic samples the model encountered were correctly classified. This tells us that the model is struggling with the subtleties of toxicity, and is only flagging messages as toxic when it is certain. This reflects the high precision and low recall seen above.

Our BERT models will be able correct these issues through their ability to recognize the intent behind messages. Refer to Table 2 for the total weighted average metrics for the NBC model. These will be a useful reference as we seek to improve performance using more complex modeling techniques.

4.2 BERT-base

The BERT class of models was introduced by Jacob Devlin and colleagues at Google in 2018 and is a strong Transformer-based language model that reads text bidirectionally. BERT-base is built from stacked Transformer encoders which contain multi-head self-attention, standard feedforward neural networks and residual connections with layer normalization.

We introduced BERT-base with a single fine tuning layer to distinguish between the different classes (Neutral, Banter, and Toxic) as a baseline to compare to our larger models. We then fine tuned the model over several epochs and measure its performance. Below is a confusion matrix representing its performance across the classes on the test set:



Class	Precision	Recall	F1-score
banter	0.90	0.83	0.87
neutral	0.73	0.82	0.77
toxic	0.55	0.46	0.50
Accuracy		0.79	
Macro Avg	0.73	0.70	0.71
Weighted Avg	0.80	0.79	0.79

Table 3: BERT-base performance, Test set

BERT-base clearly does better on this classification task than our simple Naive Bayes Classifier. Notably it performs fairly well on the neutral classification with precision of 0.73, recall of 0.82 and F1-score of 0.77. It does exceptionally well on the banter labels with a precision of 0.90, recall of 0.83 and F1-score of 0.87. However, BERT-base still struggles to understand the nuances between toxic chats and banter, this is seen in the metrics with a Precision score of 0.55, recall of 0.46 and F1-score of 0.50. Overall, while BERT-base demonstrates a strong general performance, its results still suggest that it struggles to understand the subtle nuances between toxic chats and banter.

4.3 BERT-base + LoRA

Low-Rank Adaptation (LoRA) injects small trainable rank-decomposition matrices into the attention layers of a frozen pretrained model, drastically reducing the number of trainable parameters while preserving the representational capacity of the base architecture (Mangrulkar and Paul, 2023). We applied LoRA to the same BERT-base backbone used in the previous section to test whether parameter-efficient fine-tuning could improve classification quality, particularly on the minority toxic class, without a full weight update.

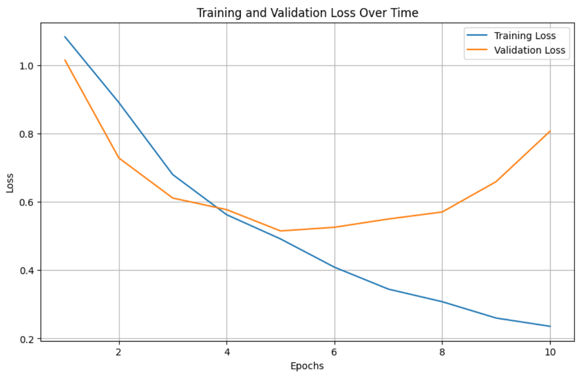
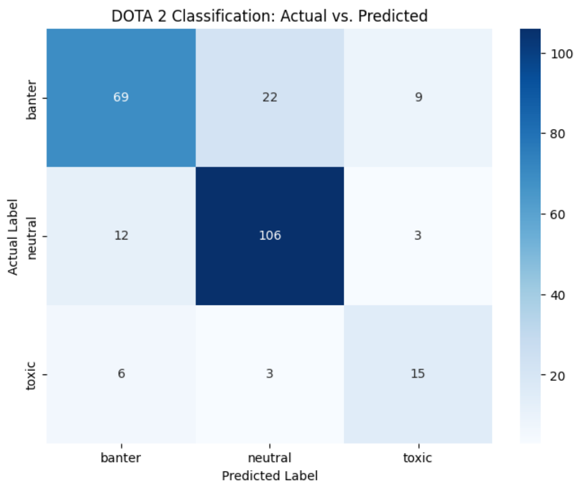


Figure 1: Training and validation loss for BERT-base + LoRA.

Figure 1 shows the training and validation loss curves over 10 epochs. Both losses decrease together through epoch 5 before diverging—training loss continues to fall while validation loss climbs, signaling overfitting. The best model checkpoint was saved at peak validation macro F1.



The most important change LoRA introduces is in toxic detection. Toxic F1 rises from 0.50 to 0.59, driven by a recall jump from 0.46 to 0.6. This meant the model caught nearly two-thirds of genuinely toxic messages instead of missing more than half. Because LoRA constrains the model to update only a small number of parameters, it acts as an implicit regularizer that prevents over-specialization on the majority classes and forces the model to attend more evenly to the sparse toxic class. Overall accuracy dips only slightly from 0.79 to 0.78, and macro F1 improves from 0.71 to 0.72. For a classifier whose primary goal is catching genuine harassment, this is a favorable tradeoff.

Class	**Precision**	**Recall**	**F1-score**
banter	0.81	0.88	0.84
neutral	0.79	0.69	0.74
toxic	0.56	0.62	0.59
Accuracy		0.78	
Macro Avg	0.72	0.73	0.72
Weighted Avg	0.78	0.78	0.77

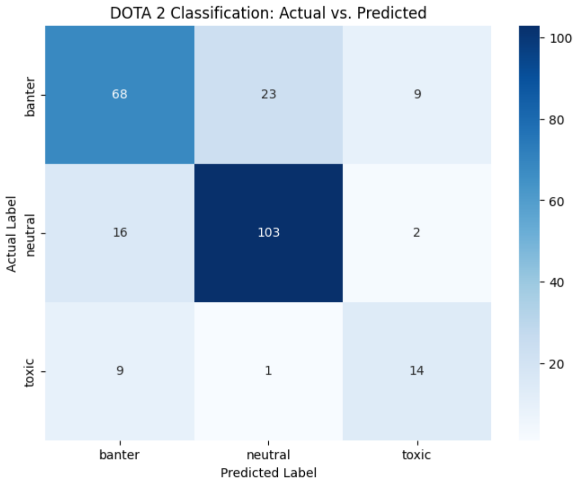
Table 4: BERT-base + LoRA performance, Test set

The confusion matrix reinforces this. Of the 9 toxic samples still misclassified, only 6 land in the banter bucket and 3 in neutral, meaning the model consistently picks up on aggressive intent even when it misjudges severity, a far less harmful failure mode than letting toxic messages pass entirely undetected. Combined with banter classification remaining strong at an F1 of 0.84, these results suggest that LoRA offers a good middle ground with better minority-class sensitivity, while not meaningfully degrading the model's handle on the majority classes.

4.4 BERT-large + LoRA

Our final model to build, train, test, and analyze is BERT-large. The architecture of BERT-large is identical to BERT-base except for the fact that BERT large has 3.09 times more parameters to tune compared to BERT base (Efimov, 2023). In our research, we discovered that more parameters can help an LLM understand complex data patterns, potentially leading to higher accuracy. (Giattino et al., 2026)

Thus, our primary aim in proceeding from the BERT-base to the BERT-large analysis is to identify the effect of parameter size when other factors are held constant. To achieve this goal, we included Low-Rank Adaptation just as in our previous model architecture, used a single fine-tuning layer, and performed over the same number of epochs. Below is a confusion matrix and a numeric table representing test set performance:



Class	**Precision**	**Recall**	**F1-score**
banter	0.81	0.85	0.83
neutral	0.73	0.68	0.70
toxic	0.56	0.58	0.57
Accuracy		0.76	
Macro Avg	0.70	0.70	0.70
Weighted Avg	0.75	0.76	0.75

Table 5: BERT-Large + LoRA performance, Test set

While BERT-Large with LoRA does show some improvements over BERT-base, improving its toxic f1-score from 0.50 to 0.57, it exhibits some similar tradeoffs that BERT-base with LoRA showed, having notably lower precision on banter examples, and lower recall on neutral instances to name a few. However, as discussed previously, for this case we believe the trade off is worth it. In this sense, BERT-Large with LoRA, exhibits the same advantages as the previous model, having good minority class sensitivity, while also maintaining its ability to manage the majority classes. An interesting point of note however, is that BERT-Large with LoRA performed marginally worse across the board as compared to its more compact counterpart, BERT-base with LoRA. We provide further analysis of why we believe this was the case in the following section.

5 Analysis & Discussion

Our results largely support the initial hypothesis derived from our research: while statistical baselines (NBC) provide a foundation for decent classification, transformer-based models (BERT) are required to capture the highly-contextual nuances of gaming communication.

As expected, the Naive Bayes Classifier performed adequately on obviously distinguishable cases, but failed significantly on the boundaries between "Banter" and "Toxic", which is integral to our purpose. Its reliance on word frequency without context meant it could not distinguish between a word used in a friendly, competitive tone versus a derogatory one. The building of this model was relatively straightforward and required no additional fine-tuning to achieve its expected results.

The BERT-base model confirmed our goal that contextual embeddings would increase predictive success. However we were surprised to find that while overall accuracy improved, the model was surprisingly inadequate in labeling toxicity, often defaulting to neutral or banter. Before trying different tuning methods to remedy this issue, we wanted to see how Parameter-Efficient Fine-Tuning might change our results.

Integrating LoRA into our BERT-base model was the next step. We found that doing so slightly lowered the overall statistical averages, but yielded a more robust distribution among class performance. Specifically, the LoRA-adapted model saw a marked improvement in identifying Toxic samples, which is what we were looking to remedy after analyzing the performance of our baseline BERT. However, we determined that some of our F1-score values were not ideal, and thus we tuned multiple parts of the model in attempt to resolve the issue. The most effective tuning we did was increasing the number of epochs just slightly in order to maximize performance without overfitting. This would prove to be a large improvement despite the marginal jumps in the statistics, as the model made major jumps in predicting the higher priority label correctly, whereas before it was struggling mightily due to sampling constraints.

While switching to BERT-Large did perform well, as we mentioned previously it has overall worse performance than the smaller BERT-base with LoRA model. We believe the reason for this has to do with the mismatch between the model's capacity and the scale and structure of our dataset. In simple terms, BERT-Large's additional 200 million parameters hold it back in this specific case. The dataset is rather small, having less than 3,000 samples in total. Furthermore, there is an extreme imbalance present in the dataset, with toxic examples making up a small fraction of the data. This leads a larger model like BERT-Large to overfit dominant patterns in the data, resulting in a failure to generalize to minority classes. This issue is further amplified by LoRA, which constrains adaptation to low-rank updates. In larger models, this limited capacity may be insufficient to meaningfully adjust all layers, resulting in under-adaptation despite the greater model size. Other limitations stem from our approach itself. The toxic class makes up roughly 10% of the data, meaning even our best model is learning from a relatively small number of toxic examples. While weighted cross entropy and LoRA helped the model attend to this class more effectively, a larger and more balanced dataset would likely improve performance further. Additionally, our annotation process relied in part on GPT4o for borderline cases, and without a formal inter annotator agreement score, there is some uncertainty in the ground truth labels themselves.

There are also broader concerns about generalizability. Training on only 2,452 samples from a single game (DOTA 2) limits how confidently we can extend these results to other gaming communities with different communicative norms. The line between toxic and banter may be far blurrier in a game like FC26, where users are calmer but still competitive, as opposed to DOTA where numerous extreme cases clearly signal toxicity. Beyond that, many games have their own mixed speech patterns and native lingo. As we mentioned earlier, we dropped non English and illegible samples to avoid noise, but in practice this is not always avoidable. What may seem illegible to the naked eye can be common speech for those who play the game. A truly robust model would need to train not just on a larger dataset, but across a wide range of communities, and even then it may struggle to classify niche examples. That is why we focus on predicting the general cases where we can, and leave room to optimize further for edge cases as more data becomes available.

Despite these constraints, the consistent improvement from NBC through BERT base + LoRA suggests that finer grained toxicity classification is a promising direction. The core challenge is that safety and competitive culture need to coexist, two seemingly contradicting ideals that make labeling difficult. Expanding with more data and broader game coverage could allow this approach to scale into a more robust moderation tool that preserves the energy and atmosphere of gaming while still keeping the space safe for everyone.

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